BRANDON CUNNANE

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EDUCATION

Expected May 2022 — **M.S. Medical Physics, 3.95 GPA**
San Diego State University (SDSU)

June 2016 — **B.S. Mechanical Engineering, 3.65 GPA**
California Polytechnic State University, San Luis Obispo (Cal Poly SLO)

RESEARCH EXPERIENCE

Oct 2020 – Present — **Graduate Student, SDSU Physics (collaboration with UC San Diego Radiology)**
Advisors: Professor Usha Sinha, Professor Shantanu Sinha

- Developed a MATLAB function to determine average muscle fiber direction from DTI data, then tracked fiber motion during contraction with dynamic VEPC imaging to analyze changes in length, strain, & angle
- Developed a MATLAB script to align muscle strain tensors in the direction of corresponding DTI eigenvectors
- Working to demonstrate feasibility of using Siemens compressed sensing 4D flow sequence for muscle strain analysis
- These projects supported three abstracts submitted to ISMRM 2022

Dec 2014 – Jun 2015 — **Research Assistant, Cal Poly SLO Mechanical Engineering**
Advisor: Professor Charles Birdsong

- Assisted graduate thesis project designing a vehicle collision detection system
- Developed a MATLAB function to automate video recordings of test runs

PROFESSIONAL EXPERIENCE

Aug 2016 – Aug 2020 — **Mechanical Engineer, MRI Safety & Quality Assurance, Abbott**

- Evaluated MRI's mechanical effects (force, torque, vibration) on pacemakers
- Measured and validated cardiac lead transfer functions used for predicting heating and voltage induced from MRI RF field
- Assessed pacemaker conformity to MRI safety standard, ISO 10974

TEACHING EXPERIENCE

Aug 2020 – Present — **Teaching Assistant**

- Instructor for 8 total sections of SDSU's Principles of Physics Lab
- Responsible for teaching, designing quizzes, grading, and office hours

Jan 2013 – Jun 2014 — **Math & Physics Tutor**
Cal Poly SLO Academic Skills Center

AWARDS

Fall 2021 — Tom Ragan Memorial Endowed Fellowship
Fall 2021 — Skolil Fund Physics Scholarship
AY 2020-21 — SDSU Master's Research Scholarship